Malhar Gudekar

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Education

University of Illinois Urbana Champaign August 2024 - May 2026
Master of Science, Information Management GPA: 3.8/4
- Coursework: Web Programming, Machine Learning & Cloud, Information Management, Information Consulting, Methods of Data Science

University of Mumbai June 2019 - May 2023
Bachelor of Engineering, Computer Science GPA: 9.25/10
- Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, Object-Oriented Programming, Machine Learning, NLP

Skills

- **Languages & DSA:** Python, SQL, Data Structures & Algorithms (Arrays, Hashing, Trees, Graphs, Dynamic Programming), LeetCode
- **Backend & Systems:** FastAPI, REST APIs, Microservices, Distributed Systems, ETL Pipelines, Kafka, PySpark
- **Databases, Cloud & Tools:** PostgreSQL, Neo4j, AWS (S3, EC2, EKS), GCP, Docker, Git

Work Experience

Business Intelligence Group August 2025 - December 2025
Technical Consultant Champaign, USA
- Built and deployed scalable, microservices-based RAG and agentic AI system using FastAPI, PostgreSQL, and vector embeddings, following MLOps/LLMOps practices for production
- Designed efficient, fault-tolerant **ETL and async pipelines** to process and index large document datasets for low-latency retrieval systems
- Improved system performance by 64% through optimized indexing, pipeline tuning, and observability tooling including logging and alerting in production
- Developed backend services enabling real-time, high-performance **AI model serving** of unstructured data across distributed systems

University of Illinois Urbana Champaign May 2025 - Present
Research Assistant - iSchool Champaign, USA
- Owned **end-to-end design, development, and integration** of cross-platform **AI backend system** ingesting wearable data for 100+ users, enabling scalable data collection and downstream analytics
- Designed and optimized **PostgreSQL schema, indexing strategies, and distributed query execution plans** for high-volume datasets, improving query performance by 38% in production workloads
- Built robust, production-ready Python generative AI pipeline using prompt engineering, retrieval, and context summarization, improving response accuracy by 43%
- Developed **asynchronous backend workflows** for real-time, fault-tolerant processing of longitudinal health data with focus on scalability

Research Assistant - CHI AI in Infodemic Management January 2025 - May 2025
- Designed and deployed highly scalable NLP and generative AI pipelines in Python with model monitoring and drift tracking, reducing processing latency by 41%
- Built distributed data ingestion pipelines using **Kafka** and **Neo4j** Streams for real-time, high-throughput processing
- Designed and developed **production-grade REST APIs using FastAPI** to serve AI models, implementing validation, monitoring, and automated deployment workflows to ensure **scalable integration** across distributed systems
- Enabled large-scale data processing across social media streams using **distributed systems and large-scale processing** principles

Swift Mobil Software Solutions Provider July 2023 - October 2023
Data Analyst Mumbai, IN
- Built interactive **Power BI** dashboards for **logistics** and **mobility KPIs**, reducing reporting time by **40%** and improving operational visibility
- Processed and optimized large-scale datasets using **Python** and **PySpark** to identify failure and performance issues in distributed systems
- Developed **SQL** and **graph queries** to map dependencies, analyze operations, detect data issues, speeding root-cause analysis by **26%**

Pscope Technologies Pvt. Ltd. January 2023 - June 2023
Data Analyst Mumbai, IN
- Developed and maintained **20+ Power BI** dashboards across Sales, Finance, and Operations for enterprise clients, leveraging **SQL** and **DAX**, enabling KPI tracking and data-driven decisions
- Automated recurring **Excel**-based workflows using **VBA**, implementing validation and scheduling logic, reducing report preparation time by **42%** and improving consistency

Stu/Dio at Illinois August 2025 - Present
Project Manager Champaign, USA
- Led cross-functional team of **6** engineers and designers to deliver **AI-driven product features** for DeepCover game, leveraging feedback and performance insights for feature prioritization
- Tracked **sprint** progress, delivery metrics, and **QA issues** through **JIRA**-based workflows, managing backlog and issue resolution

Projects

Scalable ML System for Customer Risk Prediction March 2025
UIUC Hackathon
- Built an **end-to-end machine learning system** using Python, Pandas, and LightGBM to forecast customer spending and assess credit risk from large-scale **time-series** transaction data, including data preprocessing, feature engineering, model training, and evaluation
- Designed and implemented **scalable ETL and feature engineering pipelines**, capturing behavioral trends and temporal patterns such as rolling averages and spending frequency to improve model performance
- Developed a **decision framework for credit line adjustments**, incorporating model outputs and risk thresholds to improve prediction precision by 20% and enable risk-aware classification of customers